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HARDWARE AND SOFTWARE MAINTENANCE

1) Internal Parts of a Computer (Motherboard) and Their Functions

1. CPU (Central Processing Unit)

- Function: The “brain” of the computer; it performs all calculations and executes instructions.

2. RAM (Random Access Memory)

- Function: Temporary memory that stores data being used by the CPU for fast access.

3. Chipset

- Function: Manages data flow between CPU, memory, and other peripherals.

4. BIOS/UEFI Chip

- Function: Contains firmware that starts up the system and checks hardware before handing over to the operating system.

5. CMOS Battery

- Function: Powers the BIOS memory that stores system settings like date/time.

6. Expansion Slots (PCI, PCIe)

- Function: Allow you to add extra cards like graphics cards, sound cards, or network cards.

7. SATA Ports

- Function: Connects storage devices like HDDs and SSDs.

8. Power Connectors

- Function: Supply power from the PSU (Power Supply Unit) to the motherboard and its components.

9. I/O Ports (Input/Output)

- Function: Allow external devices to connect to the system (USB, HDMI, Ethernet, etc.).

10. Heatsinks & Fans

- Function: Help cool the CPU and other components to prevent overheating.

Types of Computer Faults, Causes, and Their Solutions

1. Hardware Faults

These occur when a physical component of the computer fails. Common causes include a faulty hard drive, bad RAM, a damaged motherboard, or a failing power supply.

Solution: The faulty component should be diagnosed using hardware testing tools and then repaired or replaced.

2. Software Faults

These are related to the operating system or other software programs. They may be caused by corrupt system files, incompatible software, or viruses and malware.

Solution: Running antivirus software, reinstalling the affected program, updating drivers, or reinstalling the operating system can fix the issue.

3. Power Faults

This happens when the computer doesn't receive proper electrical power. Causes may include a defective power supply unit (PSU), loose or damaged power cables, or issues with the power outlet.

Solution: Check and secure all power connections, replace damaged cables, and if necessary, replace the PSU.

4. Overheating

Computers can overheat due to dust blocking the fans, a failing cooling system, or poor airflow inside the case.

Solution: Clean the computer internally, especially the fans and heatsinks. Ensure that the cooling system (like CPU fan or GPU fan) is working properly, and improve ventilation if needed.

5. Boot Failures

A computer may fail to boot due to corrupted BIOS settings, a missing or damaged operating system, or a faulty hard disk.

Solution: Access BIOS/UEFI to check boot settings, use recovery tools to fix the OS, or replace the hard drive if it has failed.

6. Display Issues

If the screen is blank or distorted, the problem could be a damaged monitor cable, a faulty graphics card, or a non-functioning monitor.

Solution: Test with a different cable or monitor, and if needed, replace the graphics card.

7. Peripheral Faults

These involve external devices like a keyboard, mouse, or USB drives not working properly. This could be due to port issues or missing drivers.

Solution: Try using a different port, update or reinstall device drivers, or test the peripheral on another computer.

8. Network Connectivity Problems

These faults can result from bad Ethernet cables, misconfigured network settings, or a faulty network interface card (NIC).

Solution: Replace the cables, reset network settings, or update network drivers. If the NIC is faulty, it may need to be replaced

